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RESEARCH ARTICLE

Low genetic diversity and small long-term population sizes in the spring endemic watercress darter, *Etheostoma nuchale*

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Abstract Species endemic to coldwater springs in the southeastern United States are some of the rarest and most imperiled in this region, yet little is known about their genetic composition and conservation needs. Here, microsatellite based levels of genetic diversity and estimates of effective population size (N_e) were compared between a narrow spring endemic fish, *Etheostoma nuchale*, and its widespread stream-dwelling relative, *E. swaini*. We applied several analytical methods to assess how demographic history is reflected in contemporary levels of genetic diversity for populations of *E. nuchale*. Phylogenetic analyses based on mitochondrial and nuclear DNA sequence data revealed a complex history among *E. nuchale* and *E. swaini*, but suggested ancient divergence and historic periods of isolation since colonization of spring habitats by *E. nuchale*. Populations of *E. nuchale* have levels of genetic diversity approximately one-half that of *E. swaini*, a result most likely due to founder effects and recent bottlenecks. Statistically significant F_{st} values (0.05−0.27) and STRUCTURE analyses implied high levels of differentiation among *E. nuchale* populations. Estimates of current N_e suggest relatively consistent levels across populations of *E. nuchale*, but one population may suffer from habitat degradation. We suggest that high levels of population structure and low levels of genetic diversity may be typical in other spring endemics inhabiting this region. Therefore, effective management planning for these unique species will require a detailed knowledge of the genetic and demographic history of each population.

Keywords Allelic diversity · Population structure · Effective population size · Phylogeny · Microsatellites · Spring populations

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Introduction

The southern Appalachian Highlands in the southeastern United States contains numerous groundwater-fed springs with endemic, geographically restricted species (Boschung and Mayden 2004; Mirarchi et al. 2004; Timpe et al. 2009). Within this region, coldwater springs are among the most jeopardized aquatic habitats because of impoundment, habitat alteration, and pumping for municipal drinking water and agricultural irrigation (Etnier 1997; Mirarchi et al. 2004). Although spring endemics are among the rarest and most imperiled species in this region, little is known about their genetic composition and conservation needs. In general, groundwater dependent species share several characteristics with island species in terms of their patchy distributions, levels of genetic structuring, low levels of genetic variability, and increased susceptibility to stochastic events. For example, desert pupfish in warm water springs (Duvernell and Turner 1998, 1999; Martin and Wilcox 2004) and killifish in desert oases (Bernardi et al. 2007) of the western United States and snails in the mound springs of Australia's Great Artesian Basin (Ponder 1995; Wilmer and Wilcox 2007) exhibit genetic structure among springs or spring groups that are analogous to aquatic islands in a terrestrial setting. A fundamental goal in understanding the conservation needs of island species is to determine the distribution and amount of genetic

variation within and among their populations (Frankham 1997). Levels of genetic variation in island species are often reduced compared to mainland species, resulting in higher extinction risks (Frankham 1997; Wilson et al. 2009). If genetic variation is reduced in spring endemic species, as in island species, then it is critical to understand if this reduction threatens their populations.

The watercress darter, *Etheostoma nuchale* Howell and Caldwell (Teleostei: Percidae), is a small fish restricted to four spring-pool complexes in the upper reaches of the Black Warrior River drainage in north-central Alabama (Duncan et al. 2010) (Fig. 1). Due to its restricted, urban distribution within the greater Birmingham metropolitan area, *E. nuchale* was listed as federally endangered just 5 years after its description (US Fish and Wildlife Service 1970). Top priorities listed in the watercress darter recovery plan (US Fish and Wildlife Service 1992) include determining the genetic structure among populations, correcting water quality and quantity problems, creating additional populations via translocation to new sites, and determination and long-term protection of groundwater recharge areas and preferred habitat. Several of these conservation goals have, in part, been completed. For example, Mayden et al. (2005) used allozymes to demonstrate significant genetic structuring among Roebuck (RS; Village Creek) and Thomas (TH; Valley Creek) spring populations, which they considered distinct evolutionary significant units *sensu* Waples (1995). Unfortunately, other Valley Creek populations (Glenn [GS] and Seven Springs [SS]) that may represent additional distinct units were not included in their study. Several attempts to establish new populations of *E. nuchale* have been conducted. The only successful establishment was created in Tapawingo Spring (TS; Turkey Creek) (Fig. 1) in 1988 by translocation of 200 individuals from Roebuck Spring (US Fish and Wildlife Service 1992). All *E. nuchale* populations have garnered some measure of protection through land purchase and land-owner cooperation. However, increased urban development and its effects on the spring habitat and recharge of the aquifers that feed the springs is a major concern for the persistence of *E. nuchale* populations (Stiles 2004). For example, the Roebuck Spring population recently experienced a loss of approximately 11,760 *E. nuchale* due to the removal of a small dam (Fluker et al. 2009), and Glenn Spring has experienced reduced groundwater flow relative to levels in the 1960s (Howell and Black 1976; Howell and Caldwell 1965; Stiles 2004).

Given the fragmented and isolated distribution of *E. nuchale* among spring habitats within an urban setting, there is an urgent need to elucidate appropriate management units for all springs containing this endangered species. Assessing genetic variability and structure among its constituent populations is crucial for immediate and long-term conservation planning. Although long-term persistence of imperiled species depends heavily on genetic and non-genetic factors (Avise 2004), knowledge of demographic and genetic history, revealed by neutral genetic markers, can provide clues as to how the current imperilment and

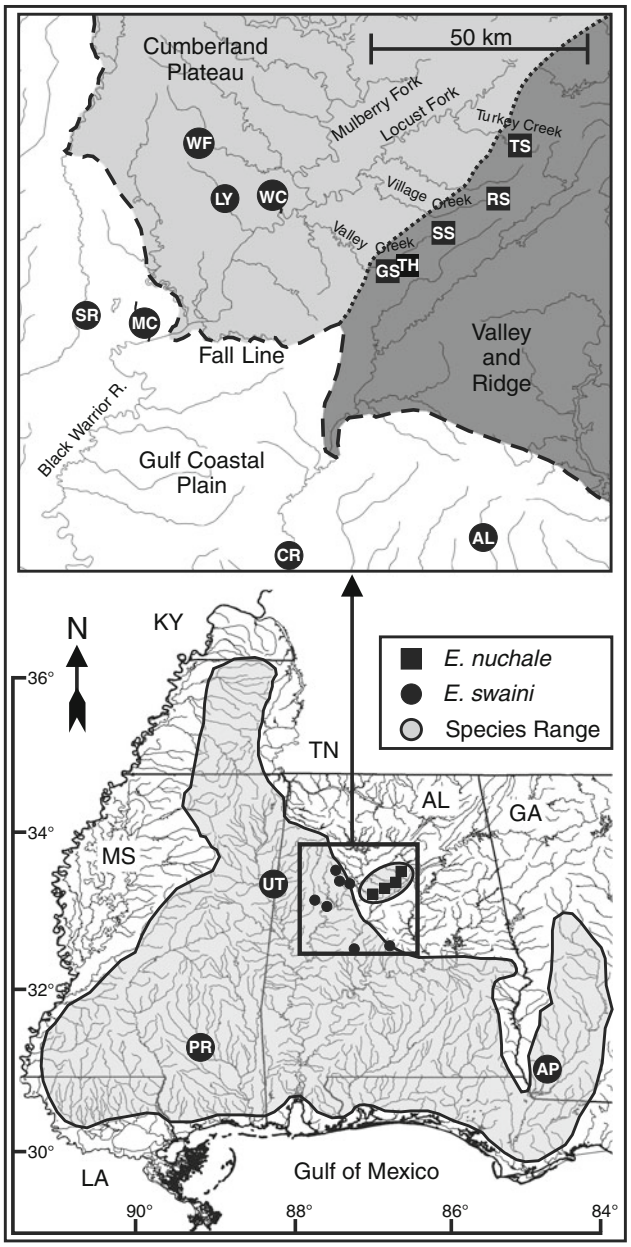

Fig. 1 Species distributions and sampling sites for *Etheostoma nuchale* (*squares*) and *E. swaini* (*circles*) used in this study (adapted from Mayden et al. 2005). Black Warrior River populations sampled: Glenn Spring (GS), Thomas Spring (TH), Seven Springs (SS), Roebuck Spring (RS), Tapawingo Spring (TS), Little Yellow Creek (LY), Mill Creek (MC), Wolf Creek (WF), and Walker County Shoal Creek (WC). Other drainages included for *E. swaini*: Alabama River (AL), Apalachicola River (AP), Cahaba River (CR), Pascagoula River (PR), Sipsey River (SR), and Upper Tombigbee River (UT). Physiographic provinces are shaded within inset

patterns of genetic variation arose (DeSalle and Amato 2004; Hurt and Hedrick 2004; Saillant et al. 2004; Sousa et al. 2008). While it is recognized that maintenance of genetic variability is important for the long-term viability and evolutionary potential of endangered species (Frankel 1974; Frankham 2005; Spielman et al. 2004), continued persistence is possible despite low levels of genetic variation (see Johnson et al. 2009). If intrapopulation genetic diversity is low and effective population sizes (N_e) are small within *E. nuchale*, management strategies may differ if reductions are recent due to anthropogenic activities versus historic, naturally low levels.

In this study, we first use phylogenetic analyses of nuclear (n) and mitochondrial (mt) DNA to examine the evolutionary history among populations of *E. nuchale* and its widespread stream dwelling sister species, *E. swaini*. This method provides a phylogenetic framework in which to better understand how the demographic history is reflected in genetic variation and effective population sizes of *E. nuchale* since diverging from its stream relative and colonizing spring habitats. Second, we use microsatellite data to further elucidate genetic structure and gene flow among populations of *E. nuchale* and *E. swaini*. Third, we use microsatellite data to test the hypothesis that contemporary levels of genetic diversity are lower in *E. nuchale* compared to closely related populations of *E. swaini*. Finally, we estimate long-term and current N_e of *E. nuchale* and *E. swaini* populations and consider whether recent or historic changes in population size can explain current genetic variability of the species and discuss how our results should guide conservation planning for *E. nuchale* and other highly imperiled spring endemics.

Materials and methods

Samples

A total of 236 individuals were collected by seine consisting of 144 *E. nuchale* from the five known spring localities and 86 *E. swaini* throughout a portion of its range (Table 1; Fig. 1). For all specimens, tissue was removed via a small fin clip. Specimens of *E. swaini* were then preserved in formalin, while those of *E. nuchale* were promptly returned to the spring. Tissues were preserved in 95% ethanol, cataloged into The University of Alabama Ichthyological Collection (UAIC), and genomic DNA was extracted using either standard cetyltrimethyl ammonium bromide-chloroform extraction procedures or the DNEasy kit (Qiagen, Valencia, CA). UAIC accession numbers for tissues used in microsatellite DNA analyses: *Etheostoma nuchale*—(TH) UAIC 14851.01; (GS) UAIC 14852.01, UAIC 14855.01, and UAIC 14857.01; (SS) UAIC 14853.01, UAIC 14858.01, and UAIC 14924.01; (RS) UAIC 14854.01; (TS) UAIC 14856.01; *E. swaini*—(Mill Creek—MC) UAIC 15138.02; (Walker County Shoal Creek—WC) UAIC 14925.01 and UAIC 15135.01; (Wolf Creek—WF) UAIC 15124.01, UAIC 15125.01, UAIC 15126.01, UAIC 15136.01, and UAIC 15137.04.

Table 1 Samples used in all genetic analyses, including outgroups for phylogenetic analyses

Species/Locality (Map ID)	*N*p	*N*m	Genbank No. (S7/ND2)
Etheostoma nuchale			
Glenn Spring (GS)	1	22	HM856115/HM856125
Thomas Spring (TH)	1	30	HM856115/HM856125
Seven Springs (SS)	1	33	HM856115/HM856126
Roebuck Spring (RS)	1	29	HM856115/HM856127
Tapawingo Spring (TS)	1	30	EF035528/EF027208
Etheostoma swaini			
Black Warrior River			
Mill Creek (MC)	1	19	HM856116/HM856128
Little Yellow Creek (LY)	1		HM856117/HM856129
Walker County Shoal Creek (WC)	1	30	HM856118/HM856130
Wolf Creek (WF)	1	30	HM856118/HM856131
Other drainages			
Alabama River (AL)	1		HM856119/HM856132
Apalachicola River (AP)	1		HM856120/HM856133
Cahaba River (CR)	1		HM856121/HM856134
Pascagoula River (PR)	1		HM856122/HM856135
Sipsey River (SR)	1		HM856123/HM856136
Upper Tombigbee River (UT)	1		HM856124/HM856137
Outgroups			
Etheostoma asprigene	1		EF035500/EF027180
Etheostoma caeruleum	1		EF035507/EF027187
Etheostoma collettei	1		EF035509/EF027189
Etheostoma ditrema	1		EF035512/EF027192
Etheostoma variatum	1		EF035550/EF027230
Percina caprodes	1		EF035498/EF027178

Number of individuals used in the phylogenetic analyses (*N*p) and microsatellite analyses (*N*m) are followed by Genbank accession numbers for DNA sequence data. Map ID corresponds to Fig. 1

DNA sequencing and microsatellite genotyping

The complete mitochondrial ND2 gene and the first intron of the nuclear ribosomal protein S7 were amplified for a subset of individuals ($N = 5$ for *E. nuchale* and $N = 10$ for *E. swaini*, Table 1) using primers and conditions listed in Lang and Mayden (2007). PCR products were purified using either the QIAquick PCR purification kit or the QIAquick gel extraction kit (Qiagen, Valencia, CA). Purified products were cycle sequenced using BigDye® Terminator v3.1 chemistry and read on an ABI PRISM®

3100 Genetic Analyzer. All base assignments were confirmed visually using Sequencher 4.2 (Gene Codes) and alignments were made by eye in BioEdit (Hall 1999). Heterozygous positions detected in S7 sequences were coded as ambiguous based on standard nomenclature of the IUPAC (Domingues et al. 2005; Ray et al. 2008).

A total of 223 specimens (144 *E. nuchale* and 79 *E. swaini*; Table 1) were genotyped for nine microsatellite loci using primers designed for a close relative, *E. caeruleum* (Eca10EPA, Eca11EPA, Eca22EPA, Eca36EPA, Eca 37EPA, Eca46EPA, Eca48EPA, Eca49EPA, and Eca71 EPA; Tonnis 2006). PCR amplifications were carried out in 12.5 µl reactions, utilizing the Failsafe™ PCR system (Epicentre Biotechnologies, Madison, WI) under the following conditions: 5–40 ng template DNA; 6.25 µl Failsafe™ PCR 2× premix; 5 µM each primer; and 0.3 U Failsafe™ enzyme mix. PCR conditions were an initial denaturation at 95°C, followed by 25 cycles of 95°C for 30 s, 56°C for 30 s, and 72°C for 40 s, followed by an additional elongation at 72°C for 2 min. Fluorescently labeled fragments were mixed with 0.5 µl GeneScan™ 500 TAMRA™ Size Standard (Applied Biosystems) and 9.0 µl Hi-Di Formamide. Genotypes were visualized on an ABI 310 Genetic analyzer (Applied Biosystems) and analyzed using GeneScan 3.7 (Applied Biosystems). Prior to statistical analyses, allele sizing was performed using the program FLEXIBIN 2.0 (Amos et al. 2007) and confirmed manually. The dataset was screened for basic typographic errors and genotyping errors due to nonamplified alleles (null alleles) and/or short allele dominance (large allele dropout) using the program MICRO-CHECKER (van Oosterhout et al. 2004).

Phylogenetic analyses: DNA sequence data

DNA sequence attributes and pairwise estimates of sequence divergence were determined in DnaSP (Rozas et al. 2003). Phylogenetic analyses were conducted independently for ND2 and S7 datasets; both included all members of the *E. asprigene* species group (*E. asprigene*, *E. caeruleum*, *E. collettei*, *E. ditrema*, *E. nuchale*, and *E. swaini*) and *E. variatum* and *Percina caprodes* as outgroups (Table 1). Maximum parsimony analyses consisted of 100 repetitions of random stepwise addition and TBR performed in PAUP*4.0b10 (Swofford 2002). MrModelTest 2.2 (Nylander 2004) was used to determine the best-fit model of sequence evolution for each of the three codon positions of the ND2 gene and for the single partition of the S7 locus. The resulting models were applied independently and used to conduct Bayesian inference of phylogeny for each locus in MrBayes v3.1.2 (Huelsenbeck and Ronquist 2001; Ronquist and Huelsenbeck 2003). Four independent runs were each conducted for 3,000,000 generations with tree sampling every 100 generations. Log-likelihood scores were plotted against generation and trees prior to stabilization were discarded as burnin. Mean posterior probability, calculated from each run (post-burnin), was used as a measure of nodal support for the 50% majority-rule consensus phylogram for each locus.

Genetic diversity and population structure: microsatellite DNA data

Conformity of genotype proportions to Hardy–Weinberg equilibrium (HWE) expectations was assessed with exact tests (Guo and Thompson 1992) and linkage disequilibrium (LD) was tested across all pairs of loci using GENEPOP version 3.1d (Raymond and Rousset 1995). Probability of significance of departure of genotype proportions from HWE expectations and LD were assessed with the Markov Chain method using 1000 dememorizations, 100 batches and 1,000 iterations per batch. Genetic diversity was measured as the mean number of alleles per locus (A), observed heterozygosity (H_o), and unbiased expected heterozygosity (H_e; Nei 1978) calculated using GENEPOP. We also calculated allelic richness (AR), an estimate of number of alleles per locus independent of sample size, using the program HP-RARE (Kalinowski 2005). The program FSTAT v 2.9.3.2 (Goudet 2001) was used to test the hypothesis that genetic diversity (based on AR, H_o, and H_e) is lower in *E. nuchale* versus *E. swaini*. Wilcoxon signed-rank tests were used to evaluate differences in AR and H_o among individual populations within species.

Population genetic structure was assessed by analysis of molecular variance (AMOVA, Excoffier et al. 1992) using three hierarchical groupings: (1) *E. nuchale* and *E. swaini*; (2) *E. nuchale* only; (3) and Valley versus Village creek populations of *E. nuchale*. AMOVA and tests for significant F_{st} values were conducted in Arlequin 3.1 (Excoffier et al. 2005) with 10,000 permutations. The program STRUCTURE 2.2 (Pritchard et al. 2000) was also used to determine K, the number of genetically differentiated clusters. This Bayesian approach uses allele frequency data to probabilistically assign individuals to groups without the need for *a priori* population designations. If an individual's genotype reflects admixture it is assigned to two or more groups, thus allowing an estimate of ancestry of an individual's genotype from one group or several groups (Falush et al. 2003; Pritchard et al. 2000). In order to determine K for *E. nuchale* and *E. swaini*, we assessed a model allowing admixture of genotypes and correlated allele frequencies between populations (Falush et al. 2003). The model was assessed with K values ranging from 2 to 10 with no prior population information and all other parameters were set to default. For each K value, we performed 10 runs with a burnin value of 100,000 and 500,000

iterations (in preliminary runs, convergence was reached after 50,000 steps). Appropriate K values were determined by ranking the estimated posterior probabilities (Pritchard et al. 2000) and confirmed using the ad hoc summary statistic ΔK (Evanno et al. 2005). Finally, we assessed migration among populations by comparing estimates of effective migrants per generation (Nm) based on M values (Slatkin 1991) estimated in Arlequin and estimates obtained using LAMARC 2.1 (Kuhner 2006) as outlined below.

N_e estimates and demographic history: microsatellite DNA data

Markov chain Monte Carlo coalescent sampling was implemented in LAMARC 2.1 (Kuhner 2006) to estimate the parameter Θ for populations of *E. nuchale* and *E. swaini*. Multi-population analyses were conducted separately for each species. Bayesian estimations were used, each with three replicates of four initial chains of 20,000 steps (1,000 trees sampled every 20 reps, burnin = 2,000) and one final chain of 600,000 steps (30,000 trees sampled every 20 reps, burnin = 30,000), with priors for Θ ranging from 1.0×10^{-5} to 10.0. We also used LAMARC to estimate immigration rates ($M = m/\mu$) from each population into each of the others, where m is the chance for a lineage to immigrate per generation, and μ is the mutation rate per site per generation (Kuhner 2006; Kuhner and Smith 2007). Priors for M ranged from 1.0×10^{-2} to 1,000. The number of effective immigrants per generation (4 Nm) for each population pair was estimated by multiplying M by its estimate of Θ for the recipient population (Kuhner 2006). Long-term N_e was estimated using the equation $\Theta = 4N_e\mu$. LAMARC based estimates of Θ and a fixed mutation rate (μ) of 5×10^{-4} mutations/locus/generation (Goldstein and Schlötterer 1999; Yue et al. 2007) were substituted into the equation above to solve for N_e. Since long-term N_e may not reflect recent demography, we used the program LD-N_e (Waples and Do 2008) to estimate current N_e based on the linkage disequilibrium method (Hill 1981) with a bias correction (Waples 2006).

We used two complementary methods to assess whether populations have experienced recent declines in effective population size. The first used the heterozygosity excess test based on the methods of Cornuet and Luikart (1996) using the program BOTTLENECK 1.2.02. Briefly, in populations that have experienced recent reductions in effective population size, allele numbers are reduced faster than gene diversity. Thus, in recently bottlenecked populations, gene diversity (*He*, Hardy–Weinberg heterozygosity) will be significantly greater than heterozygosity expected based on number of alleles and sample size (*Heq*) (Luikart and Cornuet 1998; Luikart et al. 1998; Piry et al. 1999). Wilcoxon signed-rank tests were used to assess significant deviations from stationarity between *He* and *Heq*, implementing 10,000 replicates under a two-phase model (TPM), in which 95% of mutations were single-step with a 12% variance of multi-step mutations (Piry et al. 1999). Because most microsatellites do not evolve under a strict stepwise mutation model or infinite allele model, we chose the intermediate TPM model recommended by Di Rienzo et al. (1994). The second approach involved estimating M-ratios of the number of alleles to the range in allele size and comparing them to population specific critical M values (M_c) using the programs M_P_Val.exe and Critical_M.exe (Garza and Williamson 2001). This method is thought to reflect population change over a longer period of time than BOTTLENECK methods (Garza and Williamson 2001). The M-ratio was estimated using a TPM with 90% single-step mutations, Δ_g (average size of non one-step mutations) = 3.5, and pre-bottleneck Θ values of 4 and 10 for *E. nuchale* and *E. swaini*, respectively. Because M is expected to decrease with reduced effective population size, populations with M-ratios close to one should represent a more stable population size, whereas declining populations should show lower ratios ($<\sim 0.7$) due to the rapid loss of alleles when compared to the range in allele size (Garza and Williamson 2001). Statistical significance of the observed ratio was tested using 10,000 simulations.

Results

Phylogenetic analyses

When aligned with outgroup sequences, the ND2 dataset (1,047 bp) exhibited 634 constant and 285 phylogenetically informative characters. The S7 dataset (538 bp) had 436 constant and 40 phylogenetically informative characters. Independent maximum parsimony analyses of the ND2 and S7 datasets resulted in three (902 steps) and four (122 steps) trees of equal length, respectively. MrModelTest identified the HKY + I + Γ and GTR + I + Γ models as best for S7 and ND2 sequences, respectively using the Akaike Information Criterion (Posada and Buckley 2004; Posada and Crandall 1998). Bayesian analysis of both datasets reached stationarity in less than 30,000 generations, however, the first 25% of trees were discarded as burnin to construct the 50% majority rule consensus phylograms (22,500 trees used, 7,500 discarded). Parsimony and Bayesian topologies for both datasets were highly concordant, thus we only show the Bayesian phylograms (Fig. 2). The corresponding sequence divergence estimates for focal populations are listed in Table 2. The S7 analysis recovered *E. nuchale* (clade B, all sharing identical haplotypes) as monophyletic

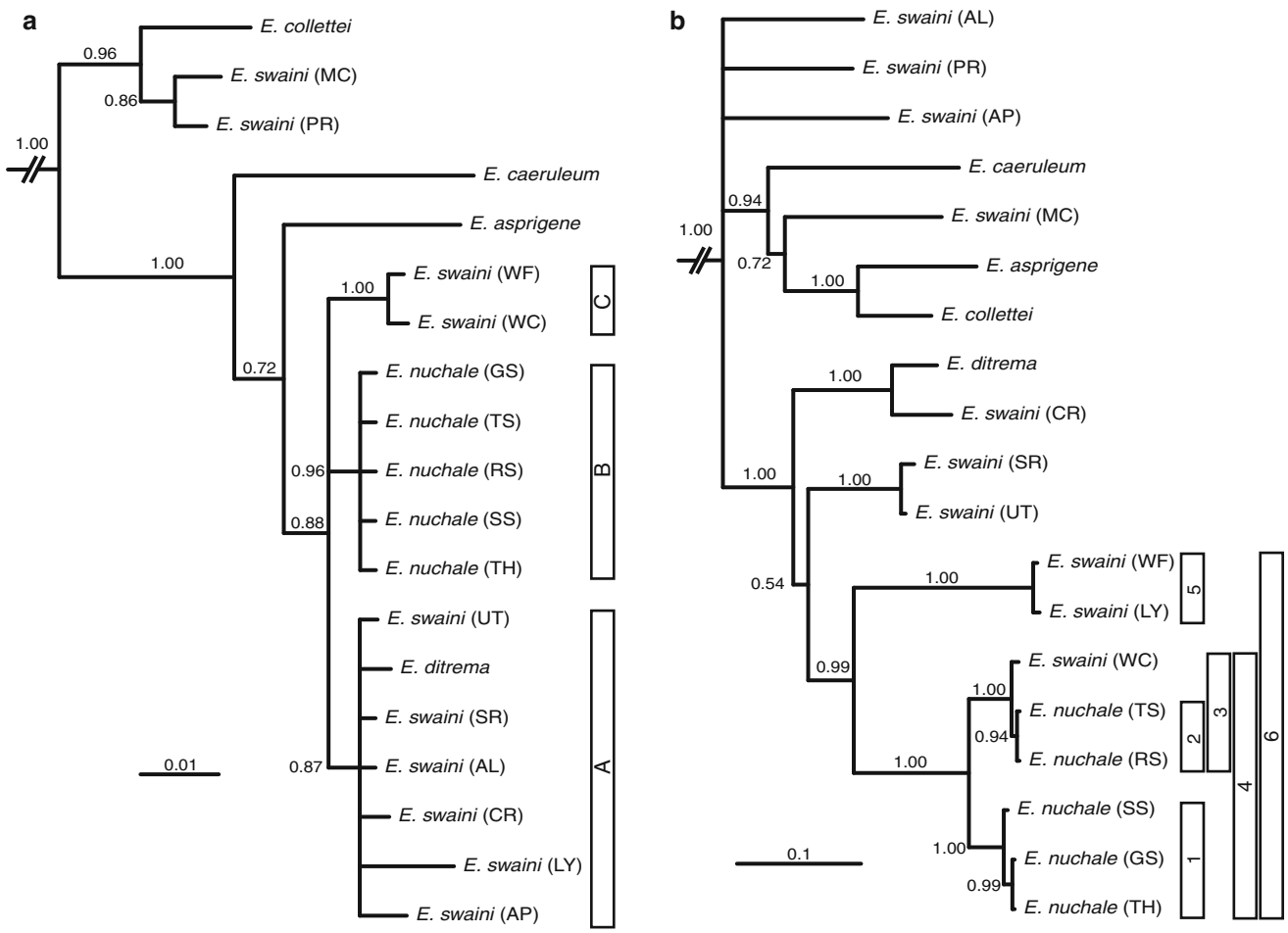

Fig. 2 Fifty percentage majority-rule consensus phylograms for Bayesian analyses of S7 (**a**) and ND2 (**b**) datasets, with mean posterior probabilities shown at nodes. Outgroup taxa *Etheostoma variatum* and *Percina caprodes* not shown

and *E. swaini* as polyphyletic; *E. nuchale* was recovered in an unresolved trichotomy with a group containing WF and WC *E. swaini* (clade C) and a weakly supported group containing *E. ditrema* plus Mobile and Apalachicola basin *E. swaini* (clade A). The ND2 phylogeny revealed a paraphyletic *E. nuchale* and a polyphyletic *E. swaini*. Glenn and Thomas spring populations of *E. nuchale* shared identical ND2 haplotypes and formed a well supported sister relationship to Seven Springs, together forming a monophyletic Valley Creek group (clade 1, 0.29% within group divergence). Roebuck and Tapawingo springs shared identical ND2 haplotypes and formed a monophyletic "Village Creek" group (clade 2) that was sister to WC *E. swaini*. Valley Creek *E. nuchale* (clade 1) formed a sister relationship to "Village Creek" *E. nuchale* + WC *E. swaini* (clade 3, 2.48% among group divergence). Clade 4

Table 2 Percentage of sequence divergence among focal populations used in microsatellite based analyses

Population	GS	TH	SS	RS	WC	WF	MC
Glenn Spring	–	0.00	0.00	0.00	0.75	0.57	3.28
Thomas Spring	0.00	–	0.00	0.00	0.75	0.57	3.28
Seven Springs	0.29	0.29	–	0.00	0.75	0.57	3.28
Roebuck Spring	2.73	2.73	2.63	–	0.75	0.57	3.28
Walker County Shoal Creek	2.82	2.82	2.73	0.29	–	0.00	3.67
Wolf Creek	9.44	9.44	9.33	9.45	9.34	–	3.49
Mill Creek	13.41	13.41	13.30	13.18	13.29	13.27	–

Estimates of S7 divergence above and ND2 below the diagonal

(*E. nuchale* + WC *E. swaini*) formed a well supported sister relationship to a group containing WF + Little Yellow Creek (LY) *E. swaini* (clade 5). Both phylogenies revealed somewhat poor resolution among other *E. swaini* and members of the *E. asprigene* species group, although in both cases, MC *E. swaini* (below the Fall Line in the Black Warrior River) was not recovered with *E. swaini* above the Fall Line in the Black Warrior River.

Genetic diversity

Amplification of all nine microsatellite loci resulted in clear, consistent fragments amenable to scoring and MICRO-CHECKER detected no typographical or genotypic errors within the dataset. All populations of *E. nuchale* were in HWE except Thomas Spring. Further examination found that two of the nine loci showed significant departures from HWE for the TH population; however, only one was significant following sequential Bonferroni correction. All populations of *E. swaini* deviated significantly from HWE. In both cases, deviations resulted from a deficiency of heterozygotes. When compared in a pairwise fashion across all loci, we found no evidence for linkage among the nine loci examined. All nine microsatellite loci were polymorphic with 3−12 and 7−21 alleles per locus across all samples of *E. nuchale* and *E. swaini*, respectively. The average number of alleles per locus ranged from 4.78 (RS) to 6.44 (GS and TH) for *E. nuchale* and 10.67 (WC) to 14.11 (WF) for *E. swaini*. Estimates of genetic diversity for all localities sampled are listed in Table 3. When compared to those of *E. swaini*, measures of AR, H_o, and H_e were significantly lower in *E. nuchale* ($P < 0.05$ for all). We found no significant differences in AR and H_o among populations within either species.

Table 3 Sample sizes and genetic diversity estimates (averaged over nine microsatellite loci) for *Etheostoma nuchale* and three populations of *E. swaini*

Species/Population	N	NA	A	AR	H_e	H_o
E. nuchale						
Glenn Spring (GS)	22	58	6.44	6.28	0.741	0.702
Thomas Spring (TH)	30	58	6.44	5.91	0.704	0.659
Seven Springs (SS)	33	49	5.44	5.04	0.700	0.667
Roebuck Spring (RS)	29	43	4.78	4.41	0.620	0.674
Tapawingo Spring (TS)	30	44	4.89	4.48	0.614	0.652
E. swaini						
Walker County Shoal Creek (WC)	30	96	10.67	9.87	0.871	0.815
Wolf Creek (WF)	30	127	14.11	12.41	0.878	0.833
Mill Creek (MC)	19	112	12.44	12.44	0.862	0.743

N = number of individuals; NA = total number of alleles; A = mean number of alleles per locus; AR = allelic richness; H_e = heterozygosity expected; H_o = heterozygosity observed

Population structure and gene flow

The AMOVA including both species suggested that 19% of the genetic diversity was distributed among populations (average $F_{st} = 0.188$; $P < 0.01$). When only considering *E. nuchale*, 24% (average $F_{st} = 0.237$; $P < 0.01$) of the molecular variation was distributed among populations, while 20% (average $F_{st} = 0.201$; $P < 0.01$) was distributed among Valley and Village creek systems. Except for the Roebuck−Tapawingo pair, pairwise F_{st} estimates (Table 4) revealed low to high levels of genetic differentiation among *E. nuchale* populations (0.051 to 0.286, $P < 0.01$ for all), while F_{st} estimates were low, yet significant among *E. swaini* populations (0.058 to 0.083, $P < 0.01$ for all). Interestingly, pairwise F_{st} estimates were greater between *E. nuchale* populations ($\bar{x} = 0.214$) when compared to estimates between *E. nuchale* and *E. swaini* ($\bar{x} = 0.178$).

The "estimated ln probability of the data" and the "mean value of alpha" in the Bayesian clustering analysis of STRUCTURE varied little among the 10 independent runs, suggesting no convergence problems in the analysis. Values of the "estimated likelihood of K" increased rapidly from $K = 2$ to $K = 6$ and showed slight increase thereafter. Similarly, the ΔK distribution showed a large peak at $K = 6$ suggesting six genetically distinct populations within our microsatellite dataset. The results revealed very high assignment accuracy for each group (0.92−0.99) (Table 5). For *E. nuchale*, individuals from Glenn and Thomas springs were assigned to group one, Seven Springs individuals were assigned to group two, and Roebuck and Tapawingo springs were both assigned to group three; each population of *E. swaini* formed a discrete group (Table 5).

N_e estimates and demographic history

Coalescent-based estimates indicated that Θ ranged from 1.75 to 2.49 for *E. nuchale* populations and 9.05 to 9.48 for *E. swaini* populations (Table 6). Estimates of long-term N_e based on LAMARC Θ values ranged from 875 to 1,245 and 4,526 to 4,738 for *E. nuchale* and *E. swaini* populations, respectively (Table 6). In comparison, current N_e estimates ranged from 49 to infinity and 284 to infinity for *E. nuchale* and *E. swaini* populations, respectively (Table 6). Bayesian LAMARC estimates of Nm were less than one in all but one *E. nuchale* population pair (into TH from GS = 2.24, into GS from TH = 1.78) and all but two *E. swaini* population pairs (into WC from WF = 1.85, into MC from WC = 4.10). Indirect estimates of gene flow

Table 4 Estimated pairwise fixation indices (F_{st} below diagonal) and effective number of migrants (*N*m above diagonal) among populations of *Etheostoma nuchale* and three populations of *E. swaini* for nine microsatellite loci

Population	GS	TH	SS	RS	TS	WC	WF	MC
Glenn Spring	–	4.641	0.952	0.706	0.681	1.491	1.588	1.138
Thomas Spring	0.051	–	0.882	0.680	0.654	1.283	1.348	1.056
Seven Springs	0.208	0.221	–	0.661	0.624	1.411	1.459	1.055
Roebuck Spring	0.261	0.269	0.274	–	1011.69	0.873	0.933	0.774
Tapawingo Spring	0.268	0.277	0.286	0.000	–	0.863	0.911	0.752
Walker County Shoal Creek	0.144	0.163	0.151	0.223	0.225	–	4.028	2.772
Wolf Creek	0.136	0.156	0.146	0.211	0.215	0.058	–	3.077
Mill Creek	0.180	0.191	0.192	0.244	0.250	0.083	0.075	–

All values significant ($P < 0.01$) excluding the Roebuck−Tapawingo comparison

Table 5 STRUCTURE group assignments and assignment probabilities for $K = 6$ for *Etheostoma nuchale* and three populations of *E. swaini*

Species/Population	STRUCTURE inferred groups					
	1	2	3	4	5	6
E. nuchale						
Glenn Spring (GS)	0.982	0.004	0.004	0.005	0.003	0.002
Thomas Spring (TH)	0.979	0.006	0.002	0.004	0.006	0.003
Seven Springs (SS)	0.004	0.981	0.004	0.004	0.003	0.003
Roebuck Spring (RS)	0.002	0.002	0.988	0.002	0.003	0.002
Tapawingo Spring (TS)	0.002	0.002	0.988	0.002	0.004	0.002
E. swaini						
Walker County Shoal Creek (WC)	0.006	0.006	0.004	0.972	0.008	0.004
Wolf Creek (WF)	0.007	0.012	0.014	0.041	0.919	0.008
Mill Creek (MC)	0.004	0.006	0.005	0.012	0.017	0.957

(*N*m; Table 4) were near zero among all but two *E. nuchale* population pairs (GS−TH = 4.64, RS−TS = 1011.69) and low among *E. swaini* populations (2.77−4.03).

Results from the BOTTLENECK analysis indicated that none of the populations for either species showed a significant excess of heterozygosity under the TPM. Alternatively, *M* values for all populations of *E. nuchale* were significantly below their respective M_c values and all but one population of *E. swaini* was significantly lower than their respective M_c values (Table 6).

Discussion

Phylogenetic incongruence

Resolving phylogenetic and taxonomic uncertainty is a crucial step in conservation planning for endangered species (Frankham et al. 2002). This study revealed that *E. swaini* is not monophyletic and exhibits complex patterns of relationships with other members of the *E. asprigene* species group, a scenario that has been considered a "complicated nexus of hypothesized introgressions and/or incomplete lineage sorting" (Lang and Mayden 2007) and

Table 6 Microsatellite based estimates of Θ calculated with the program LAMARC, the corresponding estimates of long-term effective population size (N_e), current estimates of N_e based on microsatellite loci using the linkage disequilibrium method (LD-N_e), and results from the *M*-ratio tests for each population

Population	Θ (LAMARC)		N_e (LAMARC)		LD-N_e		*M*	M_c	*P*-values
E. nuchale									
Glenn Spring (GS)	1.935	(1.744, 2.736)	968	(872, 1368)	49	(21, ∞)	0.582	0.693	0.001
Thomas Spring (TH)	2.174	(1.668, 3.262)	1087	(833, 1631)	∞	(297, ∞)	0.626	0.706	0.003
Seven Spring (SS)	2.490	(1.377, 3.519)	1245	(688, 1760)	235	(46, ∞)	0.512	0.706	0.000
Roebuck Spring (RS)	1.750	(1.315, 3.125)	875	(657, 1563)	248	(34, ∞)	0.555	0.703	0.000
E. swaini									
Walker County Shoal Creek (WC)	9.190	(7.180, 9.635)	4595	(3590, 4818)	421	(99, ∞)	0.657	0.672	0.029
Wolf Creek (WF)	9.475	(7.966, 9.837)	4738	(3983, 4919)	∞	(133, ∞)	0.741	0.672	0.298
Mill Creek (MC)	9.051	(8.151, 9.779)	4526	(4076, 4890)	284	(71, ∞)	0.629	0.642	0.029

warrants further attention beyond the scope of this study. However, our analyses indicate that *E. nuchale* is monophyletic based on nuclear S7 sequence data and paraphyletic with respect to WC *E. swaini* based on mitochondrial ND2 sequences. There are a number of reasons to explain the lack of concordance between these phylogenies. The most common reasons for species level paraphyly or polyphyly are inadequate phylogenetic information, imperfect taxonomy, interspecific hybridization, and incomplete lineage sorting of alleles (Funk and Omland 2003). We suggest that past (but not ongoing) hybridization resulting in the introgression of RS *E. nuchale* (or its ancestor) mtDNA into WC *E. swaini* is the most likely explanation for the resulting relationships based on ND2. We rule out inadequate phylogenetic information and imperfect taxonomy due to evidence from nuclear S7 and microsatellite data, which coincide with morphological differences that distinguish *E. nuchale* from *E. swaini*. The S7 phylogeny recovered a well supported, monophyletic *E. nuchale* (clade B) that was 1.03% diverged from a well supported monophyletic clade containing WF and WC populations of *E. swaini* (clade C). In addition, Bayesian assignment tests using microsatellite data revealed no ambiguous placements of either species (Table 5) and individual assignment values of RS into WC and vice versa, were all less than 0.05. The phylogenetic relationships recovered and percent sequence divergences found are hard to reconcile with incomplete lineage sorting. Based on mtDNA, WC *E. swaini* was highly divergent from WF *E. swaini* (9.3%) but not from RS *E. nuchale* (0.29%), while WC and WF *E. swaini* shared identical S7 sequences. If the WC *E. swaini* mtDNA reflected retention of an ancestral population, one would not expect such long branch lengths separating clade 3 from clade 5. Although incomplete lineage sorting cannot be completely ruled out, it is unlikely the main explanation for the close relationship among RS *E. nuchale* and WC *E. swaini* based on mtDNA. Mitochondrial introgression and replacement as a result of historical or ongoing hybridization has become increasingly recognized not only for members of the *E. asprigene* species group (Lang and Mayden 2007; Ray et al. 2008) but also among other darter species (Bossu and Near 2009; Keck and Near 2009). Thus, we conclude that the current taxonomic status of *E. nuchale* is valid.

Historical biogeography

Previous studies based on allozymes (Mayden et al. 2005) and morphology (Howell and Caldwell 1965) suggested that vicariant events associated with the geographical Fall Line (Fig. 1) resulted in peripheral isolation of the upland *E. nuchale* from the widespread Coastal Plain species *E. swaini*, a pattern of speciation common among other upper Black Warrior River endemic darters (Bart and Taylor 1999; Porter et al. 2002) and minnows of the genus *Lythrurus* (Pramuk et al. 2007). Our ND2 phylogeny supports this hypothesis, but revealed peripheral isolation of an upland group containing both *E. swaini* populations from above the Fall Line (WC, WF and LY) and *E. nuchale* (clade 6) from lowland *E. swaini*. Although relationships based on S7 are unresolved among *E. nuchale* and *E. swaini* populations, the well supported clades B and C also suggest isolation of these upland populations. The geographic placement of *E. nuchale* and upland *E. swaini* populations coincide with two very distinct underlying upland physiographies, the Valley and Ridge Province and the Cumberland Plateau, respectively (Fig. 1). Timpe et al. (2009) suggested that deep mtDNA divergence (5.18–5.58%, 4.0–4.4 Mya) among lineages of the brownback salamander (*Eurycea aquatica*) inhabiting springs in these two physiographic provinces was likely facilitated by the complex topography of this region, which is characterized by ridges separating valleys that have potentially hydrologically isolated ancient aquifers. Interestingly, average mtDNA sequence divergence between *E. nuchale* and upland *E. swaini* (9.4%) suggests a similar, ancient divergence event, revealing that *E. nuchale* has persisted in an extremely restricted geographical setting within the Valley and Ridge Province for several million years.

Genetic diversity and population structure

Analyses of microsatellite markers indicated that *E. nuchale* has significantly lower genetic diversity (AR and H_o) compared to closely (WC and WF) and distantly related (MC) populations of *E. swaini*. Our study also revealed high levels of genetic differentiation and little evidence for migration at drainage and individual population levels for *E. nuchale*. Valley and Village creek populations showed evidence of historical isolation based on high levels of mtDNA (2.48%), microsatellite ($F_{st} = 0.20$), and allozyme loci divergence ($F_{st} = 0.19$, Mayden et al. 2005). Similar levels of microsatellite divergence were found when comparing SS to GS and TH (average $F_{st} = 0.22$), but lower mtDNA divergence (0.29%) suggests a more recent connection between populations within Valley Creek. With one exception, levels of high population divergence based on F_{st} values were corroborated by groups determined in STRUCTURE. The GS and TH populations were resolved as homogenous in the STRUCTURE results, yet showed significant genetic divergence ($F_{st} = 0.05$; $P < 0.01$). The observed admixture is explained by recent population supplementation of individuals from GS to TH following an assumed reduction in TH numbers due to habitat destruction (US Fish and Wildlife Service 1992). In this case, our data represents the natural configuration of allele

frequencies in GS, while TH may be represented by a mix of its original allele frequencies and those of GS. This may also explain why TH did not conform to HWE.

Long-term effective population sizes and demographic history

The low genetic diversity for *E. nuchale* populations is in agreement with the lower long-term N_e estimates when compared to *E. swaini*. There are several potential causes for the low diversity observed in *E. nuchale*. The first is that this is simply the signature of a founder event where diversity was lost when *E. nuchale* diverged from upland *E. swaini* and populations have remained stable since. But Akst et al. (2002) suggested that ancient founding events followed by long-term stability should allow ample time for mutation to recover microsatellite diversity. Additionally, Garza and Williamson (2001) suggested *M*-ratios could recover quickly in small, stable populations. Therefore, this scenario seems unlikely for *E. nuchale*. The second possibility is that populations of *E. nuchale* diverged from its ancestor with similar levels of genetic variation and subsequently suffered reductions from population bottlenecks. Our BOTTLENECK analysis was unable to detect recent signatures of population decrease due to heterozygosity excess. However, *M*-ratios were well below their critical values for all *E. nuchale* populations suggesting somewhat recent reductions in population size. A third possibility is that *E. nuchale* underwent reductions in population size subsequent to a founding event at speciation. We propose that this may be the most likely scenario for *E. nuchale*. Our data suggest ancient divergence from upland *E. swaini* and subsequent long-term persistence of *E. nuchale* in an extremely isolated physiography. Because long-term N_e estimates reflect population size throughout a substantial portion of their evolutionary history (Hurt and Hedrick 2004), the consistently low long-term N_e of all *E. nuchale* populations suggests these values may have been historically low. Finally, the *M*-ratio analysis suggests that all *E. nuchale* populations have suffered somewhat recent reductions in N_e and have not recovered.

Conservation implications for *E. nuchale* and other spring endemics

Based on our results, *E. nuchale* should be treated as three genetically distinct populations: (1) Glenn and Thomas Springs (Halls Creek drainage); (2) Seven Springs; and (3) Roebuck Spring. If future management strategies include captive propagation and/or translocation, these patterns of genetic differentiation should be taken into account and each of these three groups should be treated independently. Ignoring these genetic factors may lead to inappropriate recovery strategies (Frankham 2005) and potential loss of the genetic distinctiveness within *E. nuchale* populations. The only exception is the introduced Tapawingo Spring population, which could serve as a surrogate for the Roebuck Spring population should supplementation ever be needed. The introduced TS population revealed that new populations of *E. nuchale* can become successfully established despite low number of founders (200) and low genetic variation. This population has been so successful that it has not only extirpated the native spring endemic rush darter (*E. phytophilum* Bart and Taylor) from Tapawingo Spring (George et al. 2009), but also has the highest relative abundance compared to all native *E. nuchale* populations (unpublished data). However, this population has only been established for 22 years and could be susceptible to population collapse as sometimes seen in introduced species (see Simberloff and Gibbons 2004). The events following the TS population translocation reveal that movement of spring endemic species outside of their native range is highly unwarranted and can potentially jeopardize native spring-dwelling species at transplant sites (George et al. 2009).

Current N_e estimates revealed that all *E. nuchale* populations except TH were below 250. The value of infinity estimated for TH may be due to the recent supplementation from GS and subsequent population growth. However, TH may simply have moderate N_e which is often difficult to distinguish from a large N_e in these types of analyses (Luikart et al. 2010). Many of the difficulties of estimating N_e are less important than the relative comparisons that can be made among populations using the same loci (Lehmann et al. 1998). In the case of *E. nuchale*, current population N_e estimates appear relatively consistent with the exception of the low values seen in GS (49), which corresponds to the loss of habitat from reduced ground-water flow. Habitat restoration and subsequent supplementation from TH may help to recover GS to comparable levels seen in other *E. nuchale* populations. Because microsatellite based estimates of N_e may be biased downward compared to estimates obtained with mtDNA and allozymes (Lehmann et al. 1998), additional markers need to be examined to corroborate our findings.

Despite their reduced genetic variation and the fact that *E. nuchale* populations have experienced reductions in N_e, they have persisted in an extremely isolated physiography since ancient divergence from upland *E. swaini*. Long-term persistence of *E. nuchale* in such a geographically restricted setting may be possible due to the relatively stable and constant spring environment. Spring habitats differ vastly in terms of chemical and thermal properties, flow regime, and stability compared to nearby or adjoining streams (Hubbs 1995). They typically remain stable in terms of temperature and volume output, devoid of top

predators, and sustain highly abundant invertebrate populations (Hubbs 1995; Teal 1957; Mirarchi et al. 2004). We also found protracted spawning periods for *E. nuchale* populations (unpublished data) which may be directly related to the constant water temperature (Howell and Caldwell 1965; Williams 1968). In Roebuck, Tapawingo, and Seven Springs, the spawning season of *E. nuchale* may be two to four times longer than the 2-month period seen in *E. swaini* (Etnier and Starnes 1993; Ruple et al. 1984).

Although reduced genetic diversity decreases evolutionary potential and increases the extinction risk of populations (Spielman et al. 2004), long-term survival of *E. nuchale* populations, as with most other spring endemic fishes (Martin and Wilcox 2004), is most directly tied to the maintenance of groundwater resources and simple habitat protection. Our results contribute to recent findings (Timpe et al. 2009) that spring endemics in the southeastern United States may exhibit high levels of population structure and low levels of genetic diversity. Therefore, effective management planning for these unique species will require a detailed knowledge of the genetic and demographic history of each population.

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